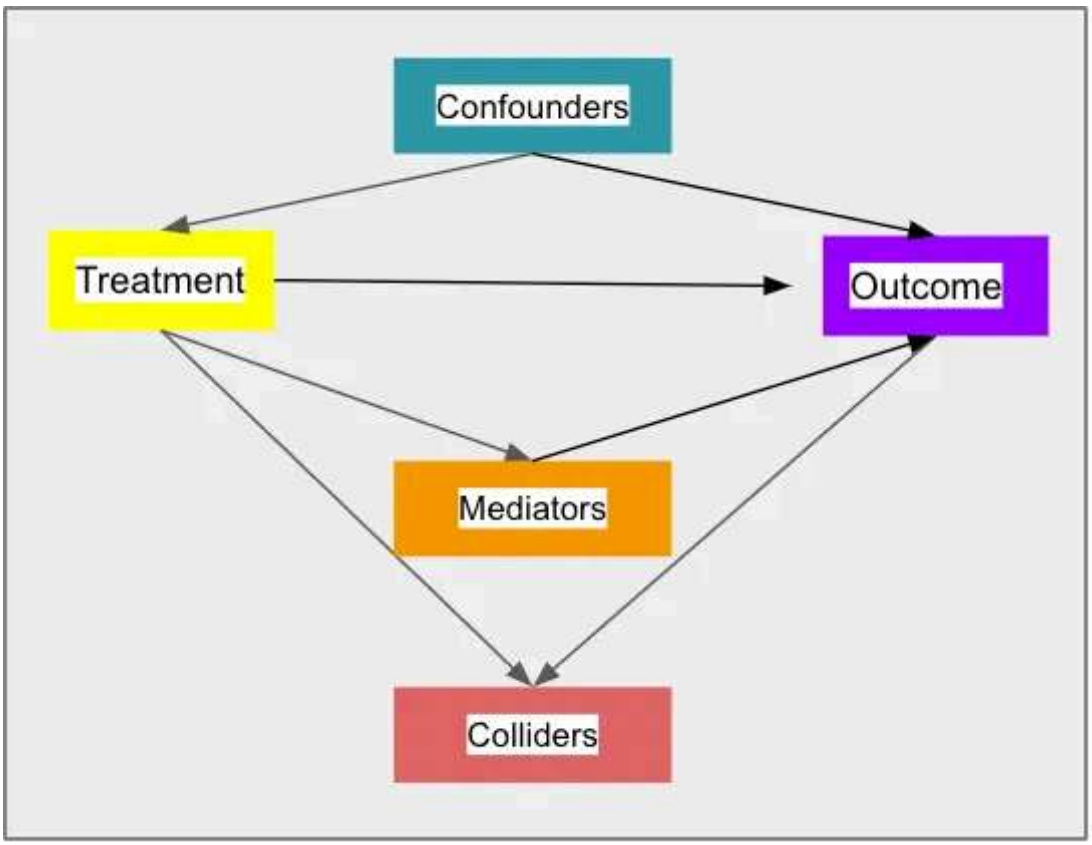
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A Simple Example to Explain Causal Graphs

Gabrielle Y Follow 8 min read · Apr 12, 2023

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This is the 4th post of my series about the big blue book “The Effect”. I really enjoy the book and would recommend everyone to read it, even just for fun.

But if you need a digest version to it, watch out for my series.

series #1: A Simple Primer on Meta Learner from 'The Effect'

series #2: Causal Inference View and Measurement for Simpson's Paradox

series #3: Causal Treatment Effect Deep Dive I — ATE

I also refer a few chapters from the causal course by Brady Neal, or papers. I will provide the links at the end.

In this post, the focus is on the causal graph: what is it, how to draw it, and how to adjust it so to answer the causal questions.

So here we go.

Causal Diagram: what is it?

> *"A causal diagram is a graphical representation of a data generating process." — 'The Effect'*

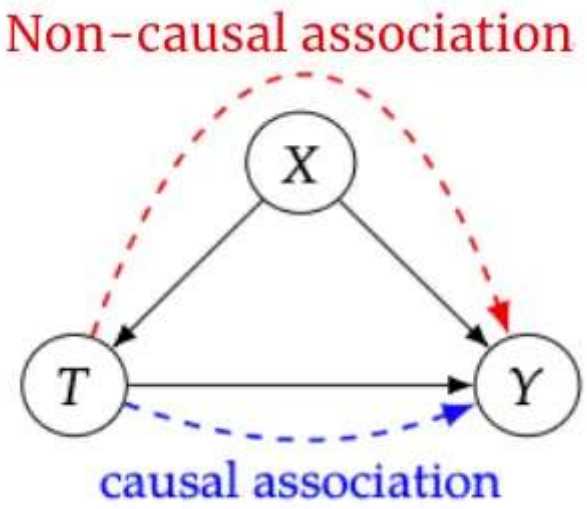
The causal diagram is consisted of node, arrows. The nodes represent the variables taken into consideration, the arrows represents the direct causal effects between the nodes.

A path is a sequence of arrows connecting two nodes regardless of the direction. There are three types of paths:

1. The chain graph: $T \rightarrow X \rightarrow Y$
2. The Fork: $T \leftarrow X \rightarrow Y$

3. The Immorality: T → X ← Y

A causal path between treatment and outcome is a path in which all arrows point away from the treatment and towards the outcome. The set of all causal paths between a treatment and an outcome comprise the total causal effect [Elwert & Winship (2014)]. Any other paths connects the treatment and outcome is non-causal paths. This is also another graphic view to explain why causal is not correlation.



Causal graph: how to draw it?

When we draft a causal diagram, we want to think about what are the possible ways the treatment and effect might be related through what variables, pick the important and relevant variables(so we won't describe the entire universe) and simply the graph. Consider an example here: say I want to understand the effect of wet soil to the bloom of flowers in spring.

Step #1: list all the possible causes to treatment and outcome, and their associations

In the data generating process of a wet soil and a flower to bloom, what leads us to observe wet soil? What are the reasons for a flower to bloom? How do the variables associated with each other? Here I list them as:

1. [treatment] wet soil
2. [outcome] flower bloom
3. sunlight hours (causes wet soil, flower bloom)
4. rain (causes wet soil, flower bloom, related to temperature)
5. water sprinkle (causes wet soil)
6. fertile land (cause flower bloom)
7. gardener visit (cause water sprinkle)
8. temperature (causes wet soil, flower bloom, water sprinkle, related to rain)
9. neighbor's sprinkle systems is on or leaking (causes wet soil)
10. prevent insects (causes flower bloom, gardener visit)

Step #2: Keep the important and relevant variables

Before we roll our sleeves and ready to draw the diagram, it is often a good idea to look at our list and understand if we keep things that matters. If we would imagine the arrow pointing in and out of the variables are likely to be tiny, we might want to remove these variables.

For example, the "neighbor's sprinkle system is leaking". This could definitely cause a wet land but might not be a super strong variable. So let's remove it.

Step #3: Draw the causal diagram and simplify it

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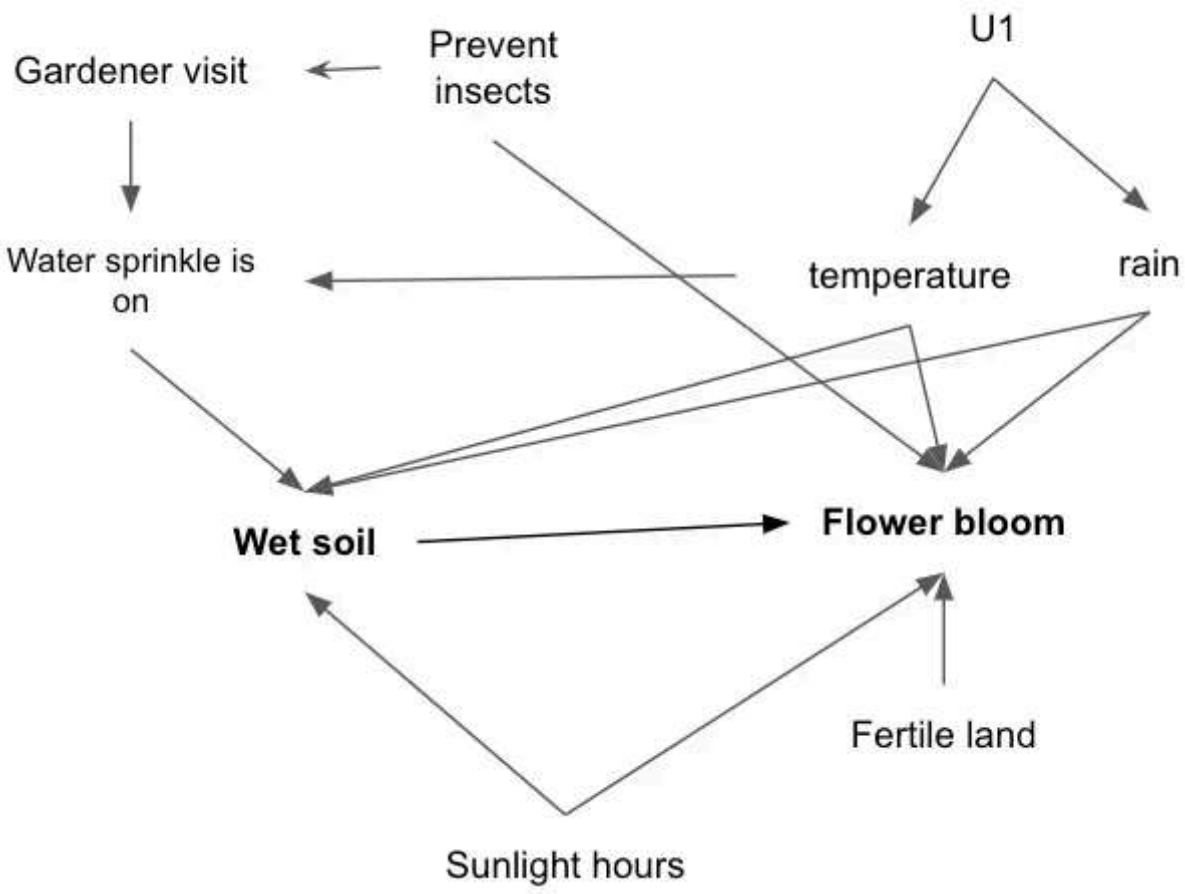
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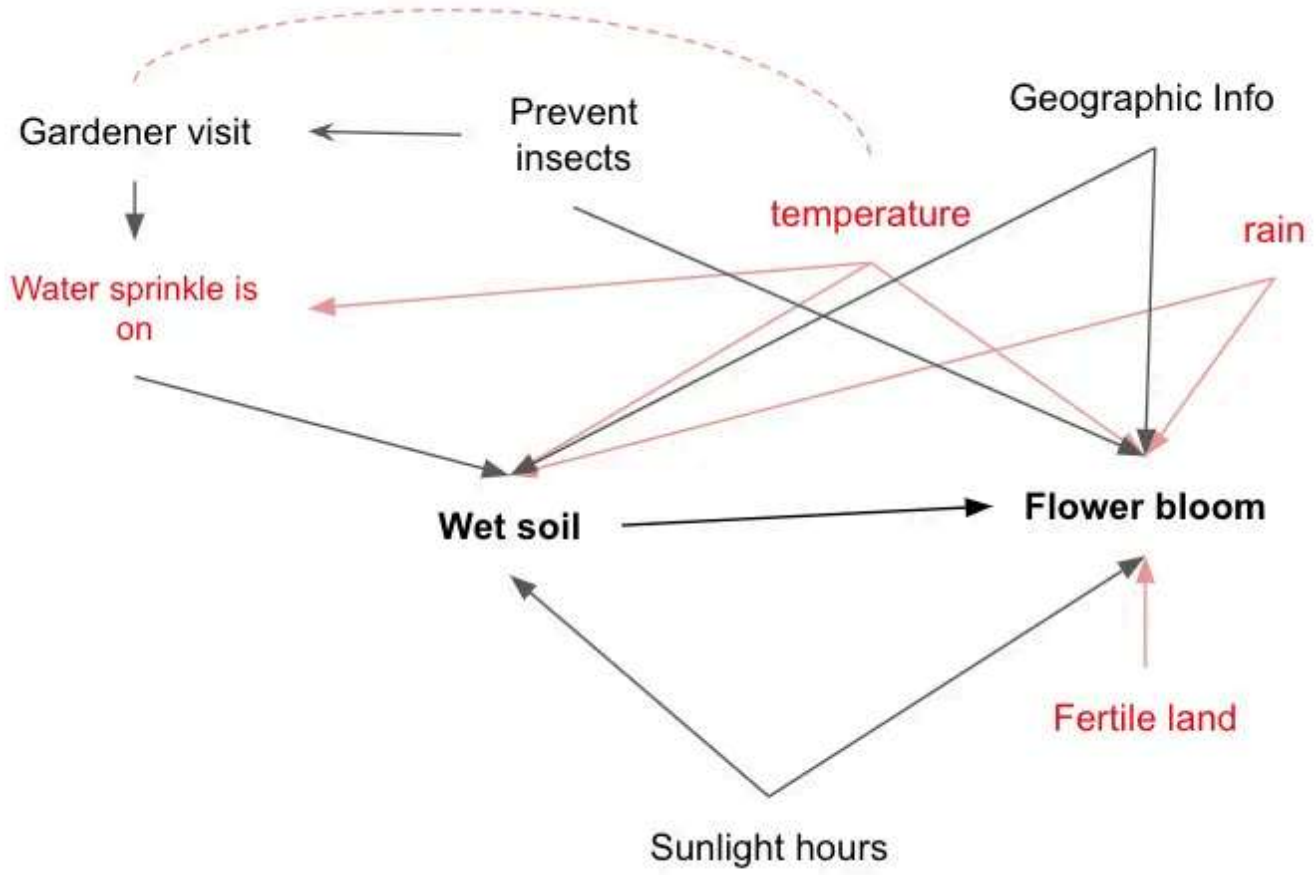
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By the above variables, the causal graph looks like below



In the below graph, I removed a few variables, which is indicated by red.



[Fertile land] I drop it because it is not on any paths between treatment and outcome. It is an important part of the data generating process but irrelevant to the question we are interested. We could include it if it reduces the prediction error but not necessary.

[Water sprinkle is on] notice this is a mediator: gardener visit will impact wet soild through water sprinkle is on. I can remove it and just have gardener visit impact wet soil directly. Another thing I notice is that the water sprinkle is on happen to be the result of gardener visit and temperature, it is also a collider: controlling on it will open a noncausal association between gardener visit and temperature, which creates collider bias.

[temperature, rain] with removal of water sprinkle is on, temperature and rain has the same set of arrows coming in and out. We can combine those to

geographic info, notice if we want to control for the geographic info, we actually control for both temperature and rain together.

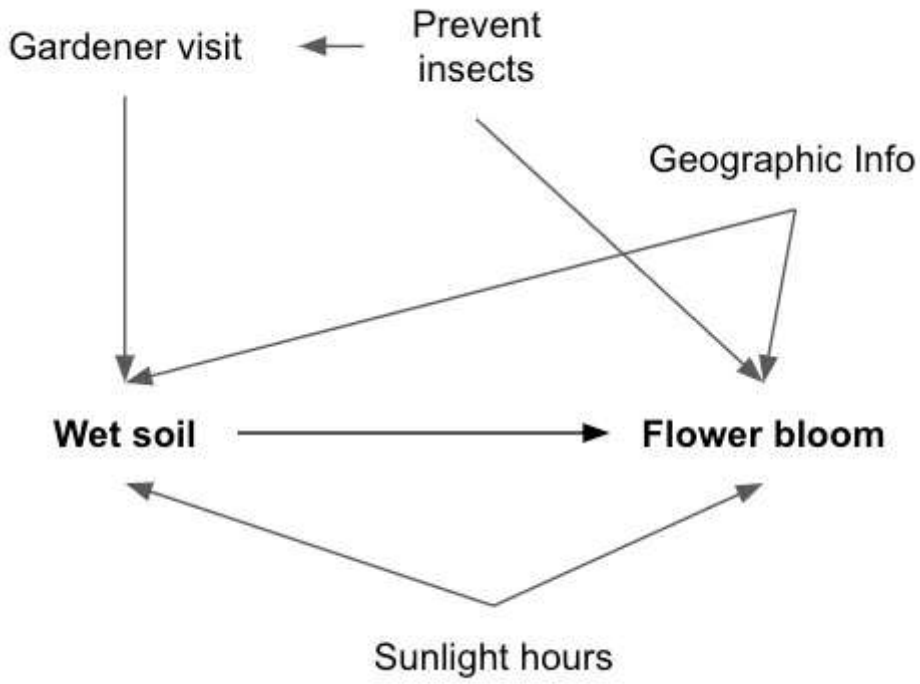
Dropping unimportant, redundant, irrelevant variables is a process of understanding how the world is working. Omitting too few would create a monster of causal diagram where variables connected with each other like a fish net; But omitting too many would also cause the omitted variable bias and ends up with a wrong identification. Where there is a variable drop from the graph, there is an assumption behind. It takes practice to get to the balance.

Adjust for causation: open the good paths, close the bad paths

Now that we have identified the causal graph for our example, we can see there are many paths that can take us from wet soil to flower bloom. But we are just interests in the direct path wet soil → flower bloom. How do we adjust other associations from data to only get to what we want? In short, we want to keep the good path and block the bad path. What are these? How do we do that?

> *"Good path: a causal path is good if it is describing a reason why treatment and outcome are related that answers your research question*
>
> *Bad path: a causal path is ad if it is describing a reason why treatment and outcome are related that is unrelated to your research question as an alternative explanation in your data. " — 'The Effect'*

First let's list all the paths between wet soil and flower bloom.



1. Wet soil → flower bloom (good path)
2. wet soil ← gardener visit ← prevent insects → flower bloom (bad path)
3. wet soil ← geographic info → flower bloom (bad path)
4. wet soil ← sunlight hours → flower bloom (bad path)

Above is an exhaust list of paths that explain why we see wet soil and flower bloom vary together. Except for path #1, all the rest of paths are alternate explanations. If we can block all the bad paths, what's left will answer our question.

So how to block the bad paths?

> ***A path between nodes T and Y is blocked by at least one variable in that path if the variable:***

> *1. is conditioned along the path, where there is a chain … → X→ …, or a fork .. … ← X → …*
>
> *2. is not conditioned, or any of its descendant are conditioned, along the path, where there is a collider … →X ← …*

Based on the above guidelines, let's control for the gardener visit, the geographic info and the sunlight hours. Then we can finally answer our question: what is the effect of wet soil on flower bloom?

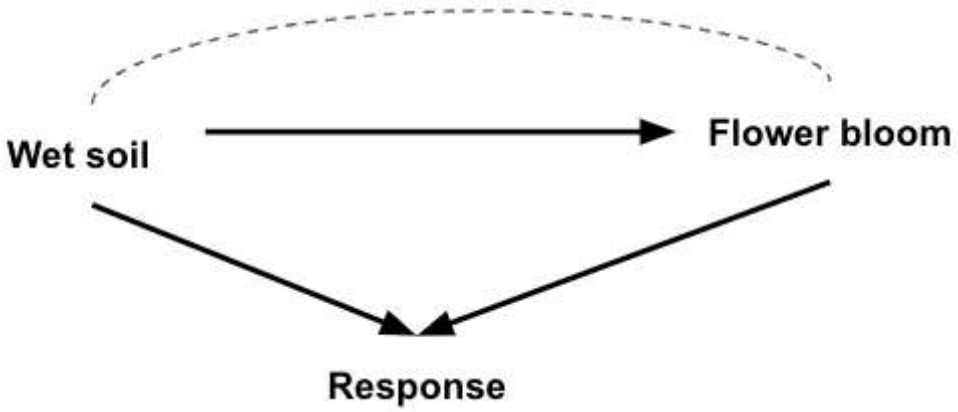
Leave the Collider untouched: a trouble when not

A collider is a variable that is a common cause of two variables, one is (or is associated with)the treatment and one that is (or is associated with) the outcome. *(Hernán et al. 2002, 2004).* Conditioning on collider will introduce a bias known as endogenous selection bias. To note, conditioning on a descendent of the collider is the same as conditioning on the collider itself as the descendent is a proxy of the collider.

I think the most absurd example of the endogenous selection bias is like, say, rain will cause the road wet, sprinkles will also cause the road wet. If we only look at the wet road data, we will probably find the rain and sprinkles are correlated. This may lack some system nuance, but the point is to showcase the logic of it. Even there is causation between the two (say, raining day has lower chance of turning on sprinkles), conditioning on the wet road will still spoil the relationship by inducing non-causal relationship.

Often, we heads down on analysis with the data we collect right away. But sometimes the data itself may already have collider bias. Take the flower example again. If, say this is a survey from a soil-selling company. People who receive and respond the survey will have some interests with their

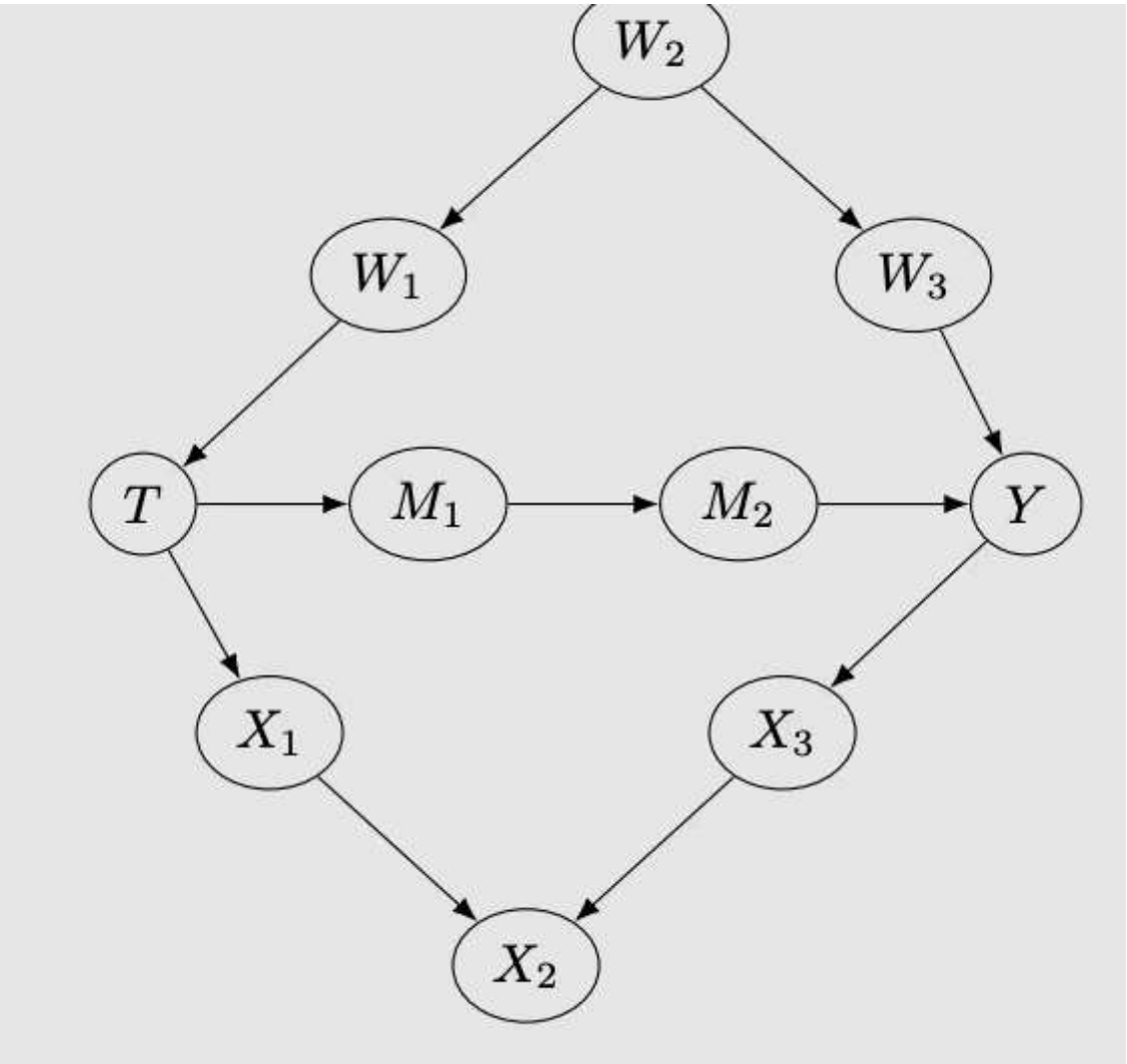
existing soil; The company may also receive survey back when flower did bloom — people usually are more motivated when their flower blooms, right? So both influence the response to the survey and responding or not then become a collider. If we analyze only the ones who send the survey back, then it is the same as control on this collider, which induce a non-causal association between wet soil and flower bloom. If there is indeed no association, then it's already bad enough. If there is a causal association, the effect will include the bias. In some extreme case where the responses are so selective, we may see negative impact between wet soil and flower bloom.



This actually is inspiring: when finish the causal graph, take a pause & think about if there is any collider bias you haven't thought about before the analysis.

Now it's your turn

Do you know now what variables to block if we want to understand the effect of T on Y?



Reference:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6089543/#S7title

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Responses (3)

Write a response

What are your thoughts?

Pablo Evia
Apr 27, 2023

What happens with the "prevent insects" variable? And I suppose that you would include "Wet soil" in your equation, right?

1 reply Reply



Tony Ike Roberts
Apr 26, 2023

Thanks for a great post! I'm curious as to what software did you use to draw your causal graphs?

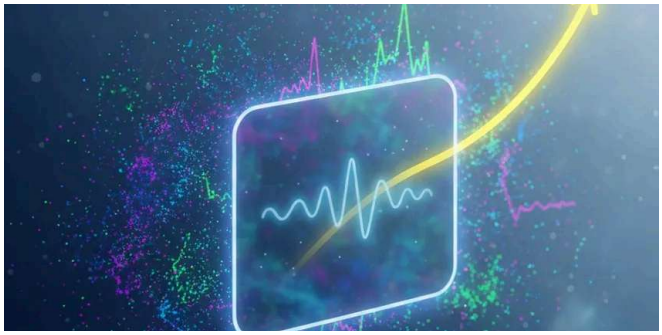
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Jeff Chen
Apr 15, 2023

It's a really great post! enjoy reading of the article a lot. Thank you

Reply

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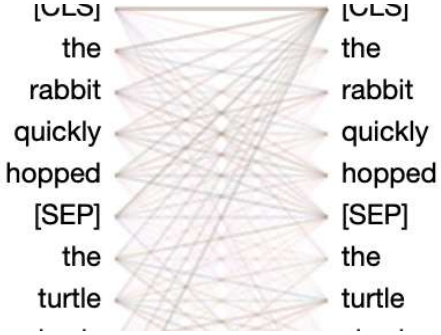


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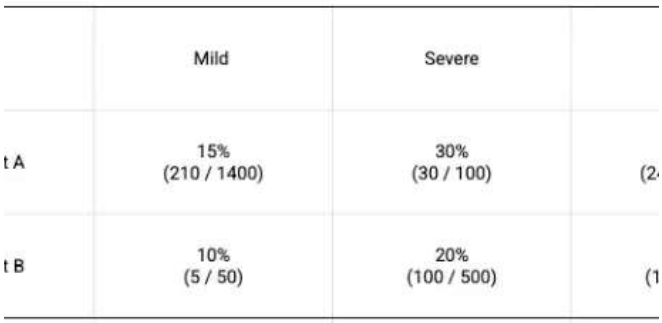


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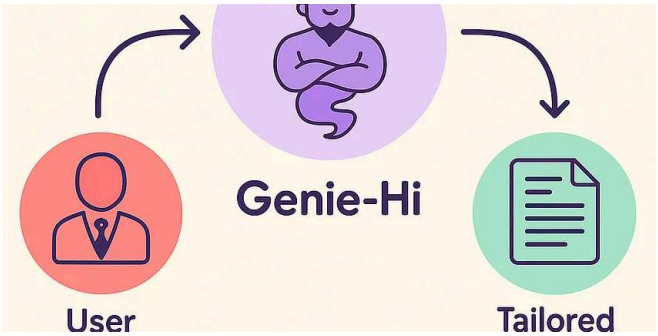
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